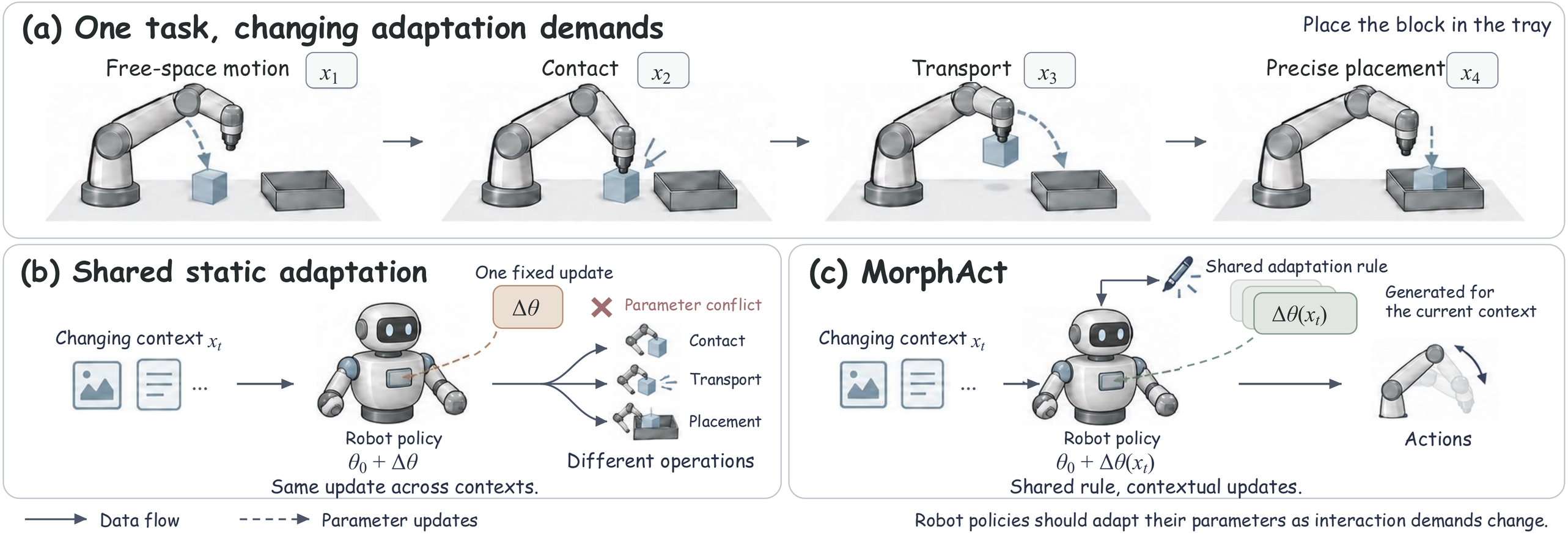

(a) One task, changing adaptation demands
Place the block in the tray
Free-space motion
x_1
Contact
x_2
Transport
x_3
Precise placement
x_4
(b) Shared static adaptation
One fixed update
$\Delta\theta$
Parameter conflict
Changing context x_t
...
Contact
Transport
Placement
Robot policy
$\theta_0 + \Delta\theta$
Different operations
Same update across contexts.
(c) MorphAct
Shared adaptation rule
$\Delta\theta(x_t)$
Generated for the current context
Changing context x_t
...
Robot policy
$\theta_0 + \Delta\theta(x_t)$
Actions
Shared rule, contextual updates.
Data flow
Parameter updates
Robot policies should adapt their parameters as interaction demands change.